

CLIPMIND AI

Video Summarization & Key Moments Detection Platform

PROJECT DOCUMENTATION

Prepared By

Satarupa Bhowmik

AI • VIDEO • NLP • INTELLIGENT CONTENT ANALYSIS

Contents.

01	Abstract
02	Introduction
03	Problem Statement
04	Objectives
05	Proposed System
06	Scope of the Project
07	Functional & Non-Functional Requirements
08	Technology Stack
09	System Architecture
10	System Workflow
11	Modules
12	Implementation
13	User Interface Screens
14	Testing
15	Results
16	Future Scope
17	Conclusion

1. Abstract

ClipMind AI is an AI-powered video analysis and summarization platform designed to help users efficiently consume long-form video content. The system automatically processes uploaded videos, converts speech into text, generates meaningful summaries, and identifies important moments with corresponding timestamps.

The platform combines video processing, speech recognition, Natural Language Processing (NLP), and Artificial Intelligence to transform lengthy videos into concise and useful information. Users can upload videos and obtain transcripts, short or detailed summaries, keywords, and important video segments through a centralized platform.

ClipMind AI supports multiple user roles, including Content Creators, Learners, Educators, and Administrators. Each role has specific permissions and functionalities. The platform also provides content analytics, upload history, activity monitoring, and AI processing capabilities.

The proposed system can be useful in education, content creation, research, business, media, and online learning environments where users need to quickly understand large amounts of video content.

2. Introduction

The rapid growth of digital video content has made video one of the most widely used sources of information and education. However, consuming long videos requires significant time and attention. Users often need to watch an entire video to locate a specific topic, important explanation, or key section.

Traditional video platforms primarily provide playback and basic search functionality. They generally do not automatically provide a complete understanding of the content without requiring users to watch the video.

ClipMind AI addresses this problem by using Artificial Intelligence to analyze video content automatically. The platform extracts spoken content from videos through speech-to-text technology and uses AI-based Natural Language Processing techniques to generate concise summaries and identify important moments.

The system aims to make video consumption faster, more accessible, and more efficient by allowing users to understand the main content of a video without watching the entire recording.

3. Problem Statement

- Users need to spend considerable time watching lengthy videos.
- Finding specific information inside a video can be difficult.
- Manual creation of video transcripts is time-consuming.
- Users may miss important moments while navigating through long videos.

- Students and educators require efficient ways to review lecture recordings.
- Content creators need faster methods for analyzing and repurposing video content.
- Traditional video platforms provide limited automated content analysis.
- Large volumes of video content are difficult to organize and understand manually.

4. Objectives

- Develop a centralized platform for video upload and processing.
- Convert video speech into accurate text transcripts.
- Generate concise and detailed AI-powered summaries.
- Identify important moments and sections within videos.
- Generate timestamps for significant video segments.
- Extract important keywords from video content.
- Provide content analytics and usage insights.
- Implement secure user authentication and role-based access control.
- Maintain video and processing history.
- Provide different functionalities for creators, learners, educators, and administrators.
- Provide a scalable architecture suitable for future expansion.
- Support cloud-based deployment and containerized application environments.

5. Proposed System

ClipMind AI proposes an intelligent video analysis platform that combines video processing and Artificial Intelligence technologies.

Processing Pipeline: Video Upload → Video Processing → Speech Recognition → Transcript Processing → AI Summarization → Key Moment Detection → Content Insights → Dashboard

The system begins when a user uploads a video. The uploaded file is validated and stored securely. The video is then processed to extract its audio and relevant information.

The extracted audio is passed to a speech-to-text model such as OpenAI Whisper to generate a transcript. The transcript is subsequently analyzed using NLP and AI models to generate summaries, extract keywords, identify topics, and detect important moments.

The generated results are stored and presented through an interactive dashboard.

Main Capabilities

- Video upload and management
- Speech-to-text transcription
- AI-powered summarization
- Key moment detection
- Timestamp generation

- Keyword extraction
- Content analytics
- User and role management
- Transcript and summary storage
- Search within transcripts
- Learning history and bookmarks
- Administrative monitoring

6. Scope of the Project

The scope of ClipMind AI covers the complete workflow from video upload to AI-generated insights.

Educational Applications

Students can use the platform to:

- Summarize lengthy lectures
- Read generated transcripts
- Find important sections of lectures
- Search for specific topics
- Bookmark useful summaries and highlights
- Review previously processed learning content

Content Creation

Content creators can:

- Upload and manage videos
- Generate transcripts
- Generate AI summaries
- Identify important moments
- Download transcripts and summaries
- Analyze content performance

Educational Institutions

Educators can:

- Upload lecture recordings
- Generate educational summaries
- Review and edit transcripts
- Share summaries with students
- Create learning materials from transcripts
- Analyze classroom content

Administrative Applications

Administrators can:

- Manage users and roles
- Monitor platform activity
- Manage uploaded content
- Monitor AI processing jobs
- View system analytics
- Manage storage and resources
- Access audit logs and reports

Future expansion can include multilingual transcription, advanced video search, recommendation systems, mobile applications, real-time processing, and additional AI models.

7. Functional & Non-Functional Requirements

Functional Requirements

User Management

- User registration and login
- Secure authentication
- Profile management
- Role-based access control
- Activity history

Video Management

- Upload videos
- Validate uploaded files
- Store videos securely
- Manage uploaded videos
- Maintain upload history

Transcript Generation

- Extract audio from videos
- Convert speech into text
- Generate transcripts
- Edit transcripts
- Store transcripts

Video Summarization

- Generate AI-powered summaries
- Generate short summaries
- Generate detailed summaries

- Extract important content

Key Moment Detection

- Identify important video segments
- Generate timestamps
- Extract highlights
- Segment content by topics

Analytics

- Analyze video content
- Generate summary reports
- Display content insights
- Track usage statistics

Administration

- Manage users and roles
- Monitor system activity
- Monitor AI processing
- Manage platform resources
- Access reports and audit logs

Non-Functional Requirements

Performance

The system should process videos efficiently and provide reasonable API and dashboard response times.

Scalability

The architecture should support increasing numbers of users, videos, and AI processing requests.

Security

Authentication, authorization, secure storage, and role-based access should protect user and content data.

Reliability

The system should handle processing failures gracefully and maintain the integrity of uploaded content and generated results.

Usability

The interface should be simple and intuitive so users can easily upload videos and access generated results.

Maintainability

The application should use modular components and clearly separated services.

Availability

The system should be designed for cloud deployment and remain accessible to authorized users.

8. Technology Stack

Layer	Technologies
	React.js, Next.js, HTML5, CSS3, JavaScript, Tailwind CSS
Backend	Python, FastAPI
	PostgreSQL, MongoDB
AI & ML	OpenAI Whisper, Hugging Face Transformers, BART, T5, TensorFlow, PyTorch
	FFmpeg, OpenCV
Authentication	JWT Authentication
	Docker, Docker Compose, AWS / Azure
Development & Testing	VS Code, Git, GitHub, Postman

9. System Architecture

ClipMind AI follows a layered architecture consisting of the user interface, API gateway, video processing and AI pipeline, analytics layer, data and storage layer, and infrastructure layer.

1. User and Role Layer

- Content Creator
- Learner
- Educator
- Administrator

2. Web/Mobile Application Layer

- Dashboard
- Video upload and viewing
- Transcript and summary viewing
- Key moment navigation
- Analytics
- Notifications
- User management

3. API Gateway

- Authentication
- API requests and routing
- Rate limiting
- Logging and monitoring

4. Video Processing and AI Pipeline

Video Upload → Video Processing → Speech Recognition → Content Summarization → Key Moment Detection → Content Insights → Output

5. Data and Storage Layer

- User data
- Video metadata
- Transcripts
- Generated summaries
- Analytics data
- Processing history

6. Infrastructure Layer

The system can be containerized using Docker and deployed using cloud platforms such as AWS or Azure.

10. System Workflow

01	User Authentication	The user registers or logs into the platform.
02	Video Upload	The authorized user uploads a video.
03	File Validation	The system validates the file format, size, and required properties.
04	Video Storage	The uploaded video is securely stored and linked to the user account.
05	Video Processing	FFmpeg/OpenCV processes the video and extracts required audio/video information.
06	Speech Recognition	Whisper converts extracted audio into a transcript.
07	Transcript Processing	The transcript is cleaned and analyzed using NLP techniques.
08	AI Summarization	AI/NLP models generate concise and detailed summaries.

09	Key Moment Detection	Important sections, topics, and timestamps are identified.
10	Content Insights	Keywords and useful content insights are extracted.
11	Result Storage	Transcripts, summaries, timestamps, and analytics are stored.
12	Dashboard Display	Users view generated results through the dashboard.

11. Modules

11.1 User Management Module

User registration, login, profile management, role-based access control, and activity history.

11.2 Content Creator Module

Upload/manage videos, generate transcripts and summaries, detect key moments, download outputs, and view analytics.

11.3 Learner Module

View videos, summaries, transcripts, key moments, search transcripts, save learning history, and bookmark content.

11.4 Educator Module

Upload lectures, generate educational summaries, edit transcripts, share summaries, create learning materials, and access classroom analytics.

11.5 Administrator Module

Manage users and roles, monitor activity and AI jobs, manage content/resources, configure settings, and access audit reports.

11.6 Video Upload Module

Video upload, file validation, secure storage, and upload history.

11.7 Transcript Generation Module

Speech-to-text conversion, transcript generation/editing, and storage.

11.8 Video Summarization Module

Short and detailed AI summaries and content abstraction.

11.9 Key Moments Detection Module

Timestamp generation, important segment identification, highlight extraction, and topic segmentation.

11.10 Analytics Dashboard Module

Video analytics, summary reports, content insights, and usage statistics.

11.11 AI Processing Module

Speech recognition, NLP summarization, keyword extraction, and content analysis.

12. Implementation

Frontend Implementation

The frontend provides an interactive interface for login, registration, dashboard, video upload, video playback, transcript viewing, summary viewing, key moment navigation, analytics, and profile management. React.js/Next.js can be used with Tailwind CSS for responsive and consistent UI design.

Backend Implementation

FastAPI provides the backend API layer and handles authentication, video upload requests, processing requests, AI processing, database operations, analytics, and permissions.

AI Processing Implementation

The AI pipeline follows: Video → Audio → Transcript → NLP Processing → Summary → Key Moments → Insights. Whisper can be used for speech-to-text conversion, while transformer-based models can support summarization and NLP analysis.

Database Implementation

The database stores user accounts, roles, video metadata, transcripts, summaries, key moments, keywords, analytics, and processing history.

Deployment

The application can be containerized using Docker and deployed to cloud infrastructure such as AWS or Azure.

13. User Interface Screens

Login & Registration

Secure account creation and authentication.

Dashboard

Overview of uploaded videos, processing status, summaries, and recent activities.

Video Upload

Interface for selecting and uploading video files.

Video Processing

Displays processing status while transcripts and AI results are generated.

Transcript Screen

Displays generated transcript and provides editing where permitted.

Summary Screen

Displays AI-generated short and detailed summaries.

Key Moments Screen

Shows important video sections with timestamps for direct navigation.

Analytics Dashboard

Displays content insights, usage statistics, and processing information.

User Management

Allows administrators to manage users, roles, permissions, and activity.

14. Testing

Functional Testing

- User registration
- Login
- Video upload
- File validation
- Transcript generation
- Summary generation
- Key moment detection
- Dashboard
- Role-based access

AI Model Testing

- Transcript accuracy
- Summary relevance
- Key moment detection accuracy
- Keyword extraction quality

Performance Testing

- Video processing time
- Upload success rate
- API response time
- Dashboard loading speed
- Concurrent processing capability
- Storage efficiency

Security Testing

- Authentication validation
- Authorization testing
- Role-based access testing
- Secure file handling
- API security testing

API Testing

- Backend API endpoints can be tested with Postman to verify request and response behavior.

15. Results

The expected outcome of ClipMind AI is a centralized platform capable of converting lengthy video content into structured and accessible information.

The system is designed to provide the following results:

- Automated video processing
- Accurate speech-to-text transcription
- AI-generated summaries
- Identification of important video moments
- Timestamp-based navigation
- Keyword extraction
- Content analytics
- Secure user management
- Role-based access
- Improved accessibility of long-form video content

Performance Evaluation

- Transcript accuracy
- Summary relevance score
- Key moment detection accuracy
- Keyword extraction quality
- Processing time
- Upload success rate
- Storage efficiency
- API response time
- Dashboard loading speed
- Concurrent processing capacity

The overall objective is to provide stable and useful results while maintaining acceptable processing and response times.

16. Future Scope

Multilingual Support

Support multiple languages for transcription and summarization.

Real-Time Processing

Provide real-time transcription and summarization for live lectures, meetings, and events.

Advanced Video Search

Allow users to search for a topic or phrase and automatically jump to the corresponding section.

AI-Based Question Answering

Provide a conversational assistant that can answer questions about uploaded video content.

Personalized Learning

Generate personalized summaries, quizzes, notes, and learning recommendations.

Mobile Application

Extend the platform to Android and iOS devices.

Advanced Analytics

Provide deeper insights into content quality, engagement, and learning patterns.

Automatic Content Generation

Generate study notes, flashcards, questions and answers, blog posts, social media snippets, and presentation content from processed videos.

Scalable Cloud Infrastructure

Optimize large-scale video processing using cloud computing, distributed processing, queues, and scalable storage.

17. Conclusion

ClipMind AI is an AI-powered video summarization and key moments detection platform designed to solve the challenge of consuming large amounts of video content efficiently.

By combining video processing, speech recognition, Natural Language Processing, and Artificial Intelligence, the platform can transform long-form videos into transcripts, concise summaries, keywords, and important timestamped sections.

The system supports different user roles, including content creators, learners, educators, and administrators, making it suitable for a wide range of applications.

The proposed architecture provides a foundation for a scalable and secure platform that can be further enhanced with multilingual support, real-time processing, conversational AI, personalized

learning, advanced analytics, and mobile applications.

Overall, ClipMind AI aims to reduce the time required to understand lengthy videos while improving accessibility, productivity, and knowledge extraction from video-based content.